

Sleep deprivation and working memory in adolescents: a randomized crossover trial

Hannah R. Okafor, Tomás de la Cruz, Mei Lin

Sleep Medicine 58 (2019) 112–120

Abstract

We examined whether one night of restricted sleep affects working memory in adolescents. Restricted sleep reduced 2-back accuracy compared with a rested night.

1. Introduction

Adolescents routinely sleep less than the eight to ten hours recommended for their age. Short sleep has been linked to poorer attention and mood, but the effect of a single short night on working memory has rarely been tested experimentally in this age group.

2. Methods

We recruited 84 students aged 14 to 17 from three secondary schools. In a randomized crossover design, each participant completed an n-back task after a normal night and after a night restricted to five hours of sleep, in counterbalanced order, one week apart. Sleep was verified by wrist actigraphy.

Accuracy on the 2-back and 3-back conditions was the primary outcome. We fitted linear mixed models with condition, order and age as fixed effects and participant as a random intercept.

3. Results

Restricted sleep reduced 2-back accuracy by 6.1 percentage points (95% CI 3.9 to 8.3; $d = 0.62$, $p < .001$). The effect on 3-back accuracy was smaller and did not reach significance ($d = 0.21$, $p = .09$). Reaction times were unchanged.

4. Discussion

A single night of five hours of sleep measurably impaired working memory in adolescents. The effect was concentrated in the moderately demanding condition, which suggests a ceiling or floor effect in the hardest condition. The sample came from three schools in one city, which limits generalisation.

5. Conclusion

Even one short night impairs working memory in adolescents, which matters during examination periods when sleep is often cut short.

References

1. Walker MP. Why we sleep. Scribner; 2017.
2. Smith J, Jones K. Sleep and memory. *J Sleep Res.* 2015;24:1–10.